Diffusion Model Assessment Report

MNIST conditional denoising diffusion model · Team midterm

ITAI 2376 · Creating Images with Diffusion Models

Part 4 — CLIP, Team Compilation, and Practical/Theoretical Assessment

Machine-generated assessment draft; author review and clean-runtime checks remain separate.

CLIP integration: recorded findings

The CLIP experiment reused 200 generated images from the
Part 3 classifier evaluation, rather than generating a different set.
Model: `openai/clip-vit-base-patch32`. Prompts and per-image results are saved in JSON/CSV.

• CLIP top-prompt/requested-class agreement: 20.50%.

• MNIST classifier/requested-class agreement on the same images: 44.00%.

• CLIP/classifier prediction agreement: 30.00%.

• Mean requested-prompt cosine similarity: 0.2985.

Figure CLIP-1 averages cosine similarity by requested generation class. Columns are candidate digit prompts; rows are requested digit classes. Strong diagonal similarity can support prompt alignment, but averaged similarity alone does not establish per-image class correctness. Cosine scores are not probabilities.

CLIP compares image-text embeddings; the separately trained MNIST classifier compares digit categories. Neither result replaces human inspection. No real-MNIST CLIP baseline or prompt-sensitivity study was run here, so differences cannot establish that CLIP is a better or worse evaluator. These observations do not prove image quality, absence of memorization, or correct diffusion implementation.

The historical Part 3 CLIP-skip outputs refer to the earlier section. This new integration is a separate Part 4 experiment, not a retroactive claim that Taki ran it.

Team compilation: results, practical evidence, and theory

Contributor labels below follow the original notebook's declared roles. They are not independently established authorship or execution records.

Contributor	Declared scope	Evidence location
Cherluk	dataset + conditional U-Net	Part 1 source, saved checks, tensor shapes and handoff
Olu	diffusion mathematics + training	Part 2 noise schedule, loss history, training source and checkpoint reference
Taki	generation + classifier evaluation	Part 3 generated images, confusion matrix and selected evaluation JSON
Michael	assembly + written assessment	Part 4 compilation and theory/practice discussion

Measured results selected for this write-up The checkpoint reference records epoch 20 and validation loss 0.02301. Final training/validation losses are 0.0235/0.0230. The MNIST evaluator has recorded real-data accuracy 97.60%, and requested-class agreement is 46.00% across 200 generated images (20 per digit). CLIP top-prompt agreement is 20.50% on 200 shared images; mean requested cosine is 0.2985.

Practical findings connected to theory 1. Forward diffusion defines known Gaussian-noise targets; the conditional U-Net learns to estimate that noise. The measured loss is therefore a noise-estimation result, not a direct measure of the requested digit appearing in the final image. 2. Reverse sampling repeatedly uses the trained noise estimate. Saved trajectories illustrate the recorded process, but a few digit-like images do not establish that every tensor pathway, schedule or checkpoint reload is correct. 3. Classifier agreement probes conditional control. CLIP adds a different, prompt-dependent image-text alignment experiment. Keep their definitions, sample counts and values separate; do not average them into a quality score. 4. Skip connections and conditioning have design rationales. Demonstrating their causal benefit requires controlled ablations; this compilation does not supply them. 5. Different generated images document variation, not proof of non-memorization. Scientific interpretation must separate measured results from explanations and untested improvement proposals.

Outstanding consistency and authorship review - Journal field most_important_finding contains a run-dependent claim requiring author review. - Journal field how_i_debugged_or_verified_it contains a run-dependent claim requiring author review. - Journal field message_to_michael contains a run-dependent claim requiring author review. - Part 4 shared-image classifier result disagrees with selected Part 3 JSON.

These findings are compiled automatically. Team members must approve the final assessment and their personal reflections. No manual sign-off is granted by this cell.

[Figure: CLIP image-text similarity]

Section 1: Introduction to Diffusion Models Our team constructed a conditional Denoising Diffusion Probabilistic Model (DDPM) to synthesize grayscale MNIST handwritten digits. The project comprises data transformations, a UNet architecture with skip connections, sinusoidal time embeddings, and conditional class embeddings. This document synthesizes our experimental findings and compares classical classifier-based evaluations with multimodal CLIP evaluations.

Section 2: Mathematical Framework and Design Decisions The forward process incrementally corrupts clean images with Gaussian noise according to a rescaled linear beta schedule from 0.001 to 0.2 over 100 timesteps. This schedule ensures that the final state achieves near-zero signal retention (alpha-bar ~ 2e-5), bridging the mathematical gap between training corruption and generation sampling. The backward process parameterizes a UNet that predicts the noise added at each step, enabling progressive generation starting from pure static.

Section 3: Empirical Evaluation and Limitations We evaluated the model using a classical convolutional neural network trained on MNIST (97.59% accuracy) and a multimodal CLIP model. Our generated requested-class agreement reached 44.00% under the classifier, with the highest fidelity on digit 1 (90.00%) and lowest on digit 6 (20.00%). The CLIP similarity scores yielded a top-prompt agreement of 20.50%, illustrating that while CLIP detects general structure, its embeddings are less calibrated for fine-grained digit classification on low-resolution grayscale datasets.